

### **1. Linear phase model**

equations only: margin, crossover, lock time

seconds

### **2. Real RTL, modelled analog**

Icarus and CocoTB, Python pump, filter, VCO

seconds to a minute

### **3. Real RTL, transistor pump and filter**

Icarus with ngspice, behavioural VCO

minutes to an hour

### **4. Whole loop at transistor level**

ngspice, started near lock, 5 to 10  $\mu$ s

hours

### **5. Cold start to lock, extracted**

ngspice overnight, typical corner

overnight